Subject: Project Assignment: Parametric Flange Base Plate Code & FEM Request

From: Engineering Design Group <design@vesselcore-industries.com>

To: Zandre (External Engineering Consultant)

Dear Contractor,

We are currently engineering a high-pressure manifold layout for our industrial boiler systems. Our internal fabrication department has generated a draft Python script to model a critical components bracket—specifically, a Parametric Structural Flange Base Plate with mounting holes.

This model must be generated programmatically inside FreeCAD so that we can modify its thickness and hole radius dynamically based on changing pressure specs.

We require you to take this baseline script, generate the 3D geometry inside FreeCAD, and perform a comprehensive Finite Element Analysis (FEA) using FreeCAD's FEM Workbench (CalculiX solver).

### Your Objectives:

1. Run the Python macro provided below inside FreeCAD to generate the baseline solid part.

2. Setup a static mechanical analysis. Fix the four bolt-hole inner faces (Constraints Fixed) and apply a uniform downward structural force of 25,000 Newtons along the Z-axis across the top face of the central hollow cylinder hub.

3. Identify structural weak points by mapping the Von Mises stress concentrations.

4. Document the peak deflection values (displacement) and suggest geometry alterations to mitigate high-stress areas (e.g., adding fillets, increasing base thickness, or modifying hole placement).

Please find our parametric macro code below.

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# FreeCAD Macro: ParametricFlangePlate.py

# Paste this directly into the FreeCAD Macro Editor and run.

```
import FreeCAD as App
import Part

def generate_flange_plate():
    # Close any existing document with the same name to avoid collisions
    doc_name = "Industrial_Flange_Plate"
    try:
        App.closeDocument(doc_name)
    except:
        pass

    doc = App.newDocument(doc_name)

    # --- DESIGN VARIATION PARAMETERS ---
    plate_width = 200.0     # Square plate dimension (mm)
    plate_thickness = 12.0  # Main plate thickness (mm)

    hub_radius = 50.0       # Central pipe outer radius (mm)
    hub_inner_r = 40.0      # Central pipe inner radius (mm)
    hub_height = 45.0       # Extrusion height of the hub (mm)

    bolt_hole_r = 10.0      # Corner mounting hole radius (mm)
```

```
hole_offset = 25.0      # Distance of bolt centers from edge (mm)

print("[INFO] Fabricating main mounting base plate...")
# Base solid plate
base_plate = Part.makeBox(plate_width, plate_width, plate_thickness)
# Center the plate on the origin (X and Y coordinates)
base_plate.translate(App.Vector(-plate_width/2.0, -plate_width/2.0, 0.0))

print("[INFO] Constructing central manifold hub cylinder...")
# Central heavy-duty connector hub
outer_hub = Part.makeCylinder(hub_radius, hub_height)
inner_hub = Part.makeCylinder(hub_inner_r, hub_height)
hollow_hub = outer_hub.cut(inner_hub)

# Fuse the base plate and the hollow cylinder together
combined_body = base_plate.fuse(hollow_hub)

print("[INFO] Boring corner structural bolt holes...")
# Define exact center points for 4 critical bolt locations
pos_x = (plate_width / 2.0) - hole_offset
pos_y = (plate_width / 2.0) - hole_offset

hole_coordinates = [
```

```
        App.Vector(pos_x, pos_y, -1.0),
        App.Vector(-pos_x, pos_y, -1.0),
        App.Vector(pos_x, -pos_y, -1.0),
        App.Vector(-pos_x, -pos_y, -1.0)
    ]

    # Drill holes through the plate layout using boolean subtraction (cut)
    for coords in hole_coordinates:
        # Make cylinder slightly taller than plate to ensure a clean cutout pass
        drill_bit = Part.makeCylinder(bolt_hole_r, plate_thickness + 2.0)
        drill_bit.translate(coords)
        combined_body = combined_body.cut(drill_bit)

    # Commit the fully compiled shape to the document tree
    structural_part = doc.addObject("Part::Feature", "Manifold_Flange_Plate")
    structural_part.Shape = combined_body

    # Recompute geometry to render model to UI window
    doc.recompute()
    print("[SUCCESS] Parametric structure generated cleanly for structural analysis.")

# Execute construction
generate_flange_plate()
```

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Once your engineering analysis is finalized, push the FreeCAD project file (`.FCStd`) alongside a detailed report outlining your recommendations to your repository. This will form a key milestone for our platform verification tracking.

Best regards,

Lead Systems Engineer

VesselCore Industries Inc.